Practice test

1. Identify the scale of measurement:
    i. Satisfaction rating
    ii. Height
    iii. Zip code
    iv. Movie ratings (e.g., 1 to 5 stars)
    v. Marital status
    vi. Country of residence

2. Given the test scores (out of 100) of 30 randomly selected students: 78, 85, 92, 67, 73, 88, 94, 81, 76, 90, 84, 69, 71, 82, 95, 77, 65, 80, 83, 68, 75, 89, 93, 86, 72, 79, 70, 87, 91, 66.
    i. Make a frequency distribution with an appropriate number of classes and class intervals.
    ii. Estimate the class midpoint for each class interval

3. A biotechnology company is conducting a study on blood sugar levels in diabetic patients as part of a clinical trial. The goal is to evaluate the impact of a new medication on blood sugar regulation. The company randomly selects 20 patients and records their fasting blood glucose levels (in mg/dL), which are as follows:
   72, 85, 90, 92, 79, 88, 77, 81, 83, 87, 80, 84, 93, 78, 82, 86, 91, 76, 74, 89
    a) Construct a frequency distribution table using a class width of 4.
    b) Using the frequency distribution table from part (a), create a histogram to visually represent the dataset.
    c) Using the frequency distribution table from part (a), construct a frequency polygon based on relative frequencies.
    d) Reconfigure the dataset into a structured stem-and-leaf display to preserve individual values while grouping them efficiently

4. Given the monthly expenditures (in dollars) on groceries for 25 randomly selected households: 250, 300, 275, 320, 290, 310, 280, 260, 350, 330, 340, 270, 300, 295, 315, 325, 285, 310, 270, 300, 280, 290, 305, 335, 310.

    i. Make a frequency distribution with an appropriate number of classes and class intervals.
    ii. Estimate relative and cumulative frequency.

5. Given the daily hours of sleep for 30 randomly selected individuals:

    7, 8, 6, 5, 9, 7, 8, 6, 7, 8, 5, 9, 6, 8, 7, 8, 7, 6, 9, 5, 6, 7, 8, 9, 6, 5, 8, 7, 6, 8.

    i. Construct the frequency distribution table.
    ii. Construct the histogram for the data.

6. Given the heights (in centimeters) of 30 randomly selected students: 160, 165, 170, 155, 175, 168, 162, 180, 172, 167, 163, 178, 169, 166, 173, 176, 158, 164, 171, 174, 161, 179, 167, 168, 172, 165, 163, 177, 159, 170.

    i. Construct the histogram for the data.
    ii. Construct the frequency polygon for the data.
    iii. Construct the ogive curve for the data

7. Given the following frequency distribution table for the heights (in inches) of students:

Height (in inches)	**Frequency**
60 - 62	5
62 - 64	8
64 - 66	12
66 - 68	10
68 - 70	5

    Calculate mean or arithmetic mean or average from this table

8. Suppose the value of a company's revenue grew by the following percentages over three years:15% in the first year, 8% in the second year and 12% in the third year. What was the average growth rate for these three years?

9. Find the median
    i. 8,5,10,2,7,3,6,4,9
    ii. 20,18,22,17,19,21,23,16,20,18
10. Suppose we have a frequency distribution table for the salaries (in thousands of dollars) of employees in a company:

Salary Range (in $000s)	Frequency
30-40	5
40-50	8
50-60	12
60-70	6
70-80	4

Find the median value.

11. Suppose we have a frequency distribution table for the ages (in years) of participants in a survey:

Age Range	Frequency
20-30	15
30-40	20
40-50	10
50-60	5
60-70	3

Find mode for this data.

12. A study was conducted by a retail analyst to assess the weekly sales performance (in hundreds of dollars) of 30 different small stores in a city. The analysis is crucial for understanding the sales distribution, identifying high-performing and low-performing

stores, and helping store managers make informed decisions about inventory and marketing strategies. The recorded data on weekly sales (in hundreds of dollars) across the stores is as follows: 57, 42, 31, 49, 68, 53, 29, 61, 50, 74, 44, 36, 70, 45, 51, 65, 39, 62, 78, 55, 48, 67, 58, 60, 66, 40, 63, 37, 41, 52.

a) Construct a Stem-and-Leaf plot for the provided data.
b) Identify the cutoff values for the lowest 10% and highest 30% of weekly sales, with proper interpretation.
c) Is there any skewness in the distribution? Provide a mathematical explanation.
d) Identify and visually represent any outliers, if present.

13. The sales of units by two shops are given by-

Sample A: 870, 750, 320, 740, 860, 930, 1060, 802, 760, 785, 608, 1206

Sample B: 586, 685, 780, 610, 830, 450, 980, 270, 640, 720, 930, 720

a) Apply the best dispersion measure on sample A and comment on your findings.
b) Which sample is more consistent?
c) Determine outlier for both samples.

14. The battery life (in hours) of two smartphone models based on real-world testing is given:

Model 1: 10, 11, 9, 12, 13, 15, 14, 13, 12, 10, 11, 14

Model 2: 8, 9, 7, 10, 11, 12, 11, 10, 9, 8, 9, 10

a) Compute and interpret the best measure of dispersion for Model 1.
b) Identify which smartphone model has more consistent battery performance.
c) Find any potential outliers in both datasets.

15. The monthly revenue (in $1000s) for two companies over a year is recorded as follows:

Company A: 120, 135, 140, 145, 160, 170, 200, 220, 185, 190, 175, 210

Company B: 90, 100, 110, 115, 140, 150, 160, 175, 155, 160, 150, 165

a) Determine the best measure of dispersion for Company A and analyze the findings.
b) Identify which company has more consistent revenue over the year.
c) Detect any outliers in both datasets.

16. Following the distance from Dhaka to world most popular cities

4827, 3619, 5073, 2145, 5294, 1432, 9761, 5582, 5314, 4950, 6406, 3298

a) Measure the Coefficient of skewness.

b) Determine variance from the data set.

c) For the given data set detect outlier/outliers.